HackTheBruh

Policy-Aware Spatial Urban Operations Compiler

An agentic planning system that turns Scottish planning policy, Edinburgh road orders, community feedback and planner goals into robust street, pedestrian and green-infrastructure designs

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Concept White Paper | Team Civilian$
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Executive Summary

HackTheBruh is an agentic urban-planning system developed for HackTheBurgh XIII. It helps planners explore Edinburgh street layouts under competing pressures: pedestrian flows, servicing and emergency access, vehicle queues, heritage and environmental protections, and uncertain demand. Instead of simulating a fixed road network alone, it proposes limited interventions, evaluates them across scenarios and explains quantitative trade-offs. This paper describes a research prototype, not an authorised council plan or legal-compliance service.

The central idea is deliberately simple: give the system a map, a set of objectives and constraints, and a demand model. HackTheBruh then constructs a mathematical representation of the city, proposes candidate interventions, evaluates those candidates at high speed on the GPU, and iterates until it finds a diverse set of robust designs. The human planner remains in control: the system presents a Pareto frontier of choices rather than declaring a single universally “perfect” city.

Edinburgh’s decisions depend on multiple sources that must not be conflated: the statutory development plan (National Planning Framework 4 and adopted City Plan 2030), the City Mobility Plan and Our Future Streets programme, traffic regulation and experimental orders, road redetermination orders, and public consultation evidence. HackTheBruh extracts source-linked records and separates (a) operative movement rules that define today’s baseline, (b) planning-policy assessments, (c) changes that would require further orders or

approvals, and (d) resident priorities. A human reviewer approves any restriction before it influences the search. All simulation figures come from mathematical tools, not the language model.

CORE OPTIMISATION	$\min_{G,x} \; E_D[J(G,x,D)] + \lambda \cdot CVaR_\alpha(J(G,x,D))$ *Optimise the network and its parameters for both average performance and bad-case risk across uncertain demand scenarios.*

What the system optimises

Vehicle movement: Reduce total travel time, queueing, bottleneck severity and delay without assuming that adding road capacity is always the correct answer.

Pedestrian movement: Model people as flows through crossings, pavements and public-space corridors, including crowding and the interaction between pedestrian priority and vehicle capacity.

Green infrastructure: Protect and improve the connectivity of parks, trees, planted corridors and other green areas, not merely the raw area of green space.

Resilience: Evaluate designs under demand shocks, closures and other disruptions, and prefer solutions whose performance degrades gracefully.

Real-time operations: Once a district is deployed, use live or simulated observations to recommend signal, routing or capacity changes rather than repeatedly redesigning the physical network.

Policy and public voice: Evaluate candidate designs against operative restrictions, identify where fresh legal orders or permissions would be required, show statutory-policy tensions, and quantify how proposals address mapped resident concerns; never label a design legally approved.

1. The Problem

Urban planning is difficult because infrastructure is coupled. A road is not an isolated object. Changing one junction changes route choice, which changes queues elsewhere. Increasing vehicle throughput at a crossing can increase pedestrian delay. Adding a green corridor can remove road capacity but create a more connected public-space network. The “best” decision therefore depends on the entire system and on what the city is trying to achieve.

Most lightweight traffic simulators answer a narrower question: “What happens to this traffic demand on this fixed network?” HackTheBruh targets the harder planning problem: “What network should we build, and how should we operate it, so that many different users and future conditions are served well?”

SYSTEM REPRESENTATION	$G = (V,E), \quad f^{car}, f^{ped}, f^{bike}, f^{transit}, \quad D \sim P(D)$ *The city is represented as a spatial graph with several interacting flows and uncertain travel demand. The prototype models car and pedestrian flows; cycling and public transport enter first as network attributes and connectivity metrics, with their own flow models left as extensions.*

2. Design Principles

Mathematics underneath, agents above — The optimisation and simulation engines should be deterministic, inspectable and testable. The AI agent orchestrates them rather than replacing them.

Grounded in policy and public voice — Objectives and constraints should not be invented by the team or guessed by a model. They are extracted from published policy and consultation evidence, quoted verbatim, and approved by the planner before any optimisation runs.

Human-readable outputs — A planner should see a map, a small number of candidate designs, the reasons for each recommendation, and explicit quantitative trade-offs.

Robustness over fragile optimisation — A network that is excellent on an average Tuesday but collapses under a road closure is not a robust planning solution.

Multi-modal by construction — Cars, pedestrians and green space are first-class variables. No mode is merely an annotation on a road network.

Hardware-agnostic experimentation — Batch independent candidate layouts and demand scenarios on available hardware. A CPU-first implementation works for the hackathon; PyTorch or CuPy GPU acceleration is optional when suitable hardware is available. No particular sponsor platform is required.

3. System Architecture

HackTheBruh is divided into six computational layers. The separation is important because it keeps the agentic portion useful without turning the project into an opaque chatbot.

Layer	Purpose
1. Policy and public voice	Planning policy, transport strategy, consultation responses and community-group publications, digested by the local model into cited constraints, objective priorities, future-demand scenarios and a located map of public concerns.
2. City state	Map geometry, junctions, roads, pavements, green areas, population or trip-demand proxies, signal plans and any operational constraints.
3. Mathematical engines	Traffic assignment, pedestrian flow, graph metrics, emissions proxies, green-connectivity metrics and stochastic scenario simulation.
4. Optimisation layer	Search over road topology, lane allocation, signal settings, crossing placement, pedestrian zones and green corridors.
5. Agent layer	A local open model turns human goals into formal objectives, chooses which tools to call, inspects results, detects weak solutions and requests further experiments.
6. Planner interface	A map and ranked-by-trade-off candidate set. The interface exposes the objectives, lets the planner alter their relative importance, and shows the policy clauses and public concerns behind each recommendation.

4. Agentic Intelligence

The agent is an auditable experiment manager. A user might ask: “Improve pedestrian movement on Lawnmarket and Castlehill, retain accessible and servicing access consistent with the currently operative order, avoid degrading the surrounding network, and flag any approvals needed for a proposed change.” It

translates that request into objectives, baseline restrictions, design scenarios and explicit assumptions. It cannot authorise a change to road rights or grant consent.

The language model does two distinct jobs. Before any optimisation, it acts as a reader: it works through policy documents and public feedback that would take a person days to cross-reference, and produces structured, cited inputs (Section 5). During optimisation, it acts as an experimenter: it decides what to test, critiques the results and explains the outcome in the language of the policies and concerns it started from. In both roles it never produces a number itself; numbers come only from the mathematical engines.

Agent loop

Agent step	Action
Read	Digest policy documents and consultation feedback into cited constraints, priorities and located concerns; ask the planner to confirm them.
Observe	Read the current network, demand summary, constraints and previous experiments.
Formalise	Translate natural language goals into objective functions, constraints and scenario definitions.
Propose	Generate one or more network modifications or operating strategies.
Simulate	Run traffic, pedestrian and infrastructure models under many demand scenarios.
Critique	Inspect the results for bottlenecks, unfair deterioration, fragility and dominated candidate solutions.
Iterate	Change the search region or objective emphasis and run another experiment.
Explain	Return a concise, inspectable account of what changed, why the candidate is on the Pareto frontier, and which policies and public concerns it addresses or strains.
AGENT DECISION	$a_t = \pi_\theta(s_t, g_t), \quad s_{t+1}=\text{Tool}(a_t,s_t)$ *π is the learned policy or planner; the tool state changes only through explicit simulation and optimisation actions. This preserves an auditable chain of computation.*

5. Policy and Public-Voice Intelligence

A mathematically attractive street layout may be inconsistent with legal traffic restrictions, damage a designated historic setting, obstruct essential access or conflict with the council's transport priorities. Edinburgh's adopted City Plan 2030 (7 November 2024) and Scotland's National Planning Framework 4 (13 February 2023) form the statutory development plan; the City Mobility Plan (updated February 2024), Our Future Streets and local consultation material provide additional strategy and evidence. Road traffic restrictions and rights of passage are governed through separate legal processes. HackTheBruh therefore reads these sources as different classes of authority, not one undifferentiated set of "rules".

5.1 What goes in

Source	What it contains	What HackTheBruh extracts
Edinburgh TROs, ETROs, temporary orders and order schedules	Movement and parking restrictions, one-way rules, banned turns, vehicle classes, time windows, exemptions, amendments and commencement dates.	Verified legal baseline by edge/turn/mode/time; unresolved exemptions and order status placed in a human-review queue.
Adopted City Plan 2030 + Scotland's NPF4 (statutory development plan)	Edinburgh spatial allocations and mapped designations; NPF4 Policies 7, 13, 15 and 20; Scottish planning decision framework.	Cited policy-alignment tests, mapped safeguards, and location-based scenarios; no invented blanket legal prohibitions.
Roads (Scotland) Act 1984 and Edinburgh redetermination orders (RSOs)	Changes to means of public passage between carriageways, footways and cycle tracks; related approved orders.	Approval-required flags for proposed rights-of-passage changes; validated baseline road-use classifications.
City Mobility Plan 2021–2030 (2024 update), Our Future Streets, implementation plans	Edinburgh priorities for people, movement and place; walking, wheeling, cycling, public transport, servicing and safety.	Objective-weight priors, access tests and named scenario objectives, not statutory prohibitions.
City of Edinburgh Council Consultation Hub and Old Town Streets consultation	Published consultation summaries and scheme-specific concerns about streets, access, disability, loading and traffic.	Geolocated public concerns, deduplicated where possible; counts and representativeness cautions visible.
Community and stakeholder publications (e.g. Spokes; accessibility and business groups)	Mode-specific evidence on cycling gaps, pedestrian safety, disabled access, deliveries, heritage and local severance.	Attributed concern themes and testable interventions, not automatically binding legal rules.
The planner's own brief	Goals, budget limits and things that cannot change.	Additional constraints and the final say over everything above.

Use lawfully available, openly published material subject to licence and access conditions. Record source URL, publisher, document version, issue date, effective date, order status, quoted span and date fetched. Remove names and other identifying details from consultation records and retain only necessary themes and locations. A local model can keep processing on-device; if cloud inference is used, use only material permitted by applicable data-handling rules and document that configuration.

Legal caveat: NPF4 and City Plan 2030 guide planning decisions under the Scottish plan-led system; they do not automatically prohibit every possible road intervention. Planning judgement and material considerations may be relevant. By contrast, currently operative traffic orders set rules for using roads, while changing a road between carriageway, footway and cycle track may need a redetermination order under section 152(2) of

the Roads (Scotland) Act 1984. The prototype reports legal status, policy alignment, and approval triggers separately; none is an automated legal determination.

5.2 From documents to mathematics

The pipeline is retrieval and schema-validated extraction, not free-form summarisation. Every record contains a verbatim source span, exact reference, source type, effective-date/status fields and confidence or abstention. Statutory development-plan policy, council strategy, operative order, proposed order and resident comment are explicitly different record classes.

Stage	What happens
Ingest	Parse City Plan 2030/NPF4, Council transport plans, signed or published orders and consultations; retain document identity, exact span, order schedule, issuer, status and date.
Triage	Route spans to: operative road rule; proposed/experimental/temporary rule; statutory planning policy; council strategy; mapped designation; consultation evidence; abstain.
Retrieve	Embed the sections; for the chosen district and topic, pull the passages that mention its streets, designations or transport modes.
Extract	Emit one typed record per rule/policy/comment: verbatim quote, section/schedule, source URL, authority, version, dates, mode, exemptions, location, legal role and uncertainty.
Classify	A mandatory phrase in a traffic order may define a baseline road restriction once in force. Development-plan policy is assessed contextually, strategy sets priorities, and consultation remains a distinct concern signal.
Ground	Match street names, junctions and designations to nodes and edges of the city graph by deterministic string and geometric matching. The model chooses from the enumerated candidates or abstains; it never writes an edge identifier itself.
Verify	Verify literal quotes and scope, inspect operative order dates/amendments/exemptions, and require planner review. Any uncertain legal effect stays UNRESOLVED, not silently enforced.
Compile	Compile two outputs: baseline legal access by mode/time/exemption and proposal-level policy/approval flags; plus objective weights, demand scenarios and located concerns.

POLICY COMPILATION	$c \mapsto$ (source_type, authority, status, dates, scope, predicate, review_flag). Baseline legal graph $A_{current}$ = {movements satisfying verified in-force orders}; candidate designs retain separate planning assessment and approval flags. Never infer legal approval from model feasibility.

WEIGHT PRIOR	w_initial = w_planner ⊙ (1 + η·p_strategy), where p_strategy reflects cited City Mobility Plan and NPF4 policy themes. The planner can revise non-binding priorities; statutory provisions are not converted into arbitrary scores.

5.3 Public concerns as a spatial signal

Public feedback is a located signal, not a plebiscite. HackTheBruh reads published, licensed Edinburgh Consultation Hub responses, Old Town Streets consultation summaries and stakeholder publications, and tags concerns such as crossing delay, accessibility, rat-running, delivery access, road safety, cycling gaps and air quality. It grounds usable locations to the graph, deduplicates repeated material, reports counts, and displays a thematic heatmap without inferring residents' identities.

CONCERN FIELD	$s_e^j = \Sigma_i \, sev_i \cdot K(dist(e, loc_i)) \cdot 1[theme_i = j]$ *Comment i has location loc_i, severity sev_i and theme theme_i; j indexes themes. The distance kernel K spreads its influence over nearby edges e.*

Each theme is paired with the model output that measures it: crossing-delay complaints with modelled pedestrian delay, rat-running with flow on residential edges, air quality with the emissions proxy. A design scores well on public voice when it improves those quantities in the places where people raised them. Cycling concerns, which the prototype has no flow model for, are matched to a cheap graph metric instead: the share of origin-destination pairs joined by a continuous cycle-suitable route. Themes with no measurable counterpart, such as parking, are shown on the heatmap but left out of J_voice rather than scored with an invented proxy.

PUBLIC-VOICE OBJECTIVE	$J_{voice}(x) = \Sigma_j \, \Sigma_e \, s_e^j \cdot m_e^j(x)$ *$m_e^j(x)$ is the modelled metric matched to theme j on edge e under design x, divided by its baseline value so that themes are comparable. J_voice joins the Pareto vector alongside J_car, J_ped, J_green and J_emit.*

Consultation respondents are self-selected and are not a representative sample of the city. HackTheBruh therefore keeps public voice as its own visible axis rather than silently folding it into a single score, reports how many comments sit behind each hotspot, and leaves the weighting to the planner.

5.4 Policy check and explanation

Every candidate receives a transparent decision-support check with four independent outputs: (1) consistency with known operative road restrictions in the chosen scenario, (2) additional TRO/ETRO/redetermination/other approvals that may be needed, (3) possible support or tension with NPF4, City Plan 2030 and mobility strategy, and (4) local public concerns addressed or worsened. A typical explanation: "This option reduces modelled pedestrian delay on Lawnmarket but diverts vehicle flow to surrounding streets and would require separate authority review before implementation." It must never assert that a prototype design is legally approved.

5.5 The constraint compiler: from regulation to the feasible set

Section 7 searches over hypothetical interventions to Edinburgh's roads. There is a crucial distinction between the legal baseline and whether an intervention could be proposed: an existing traffic prohibition removes a movement from the current admissible route graph, but does not mean the physical street can never be redesigned. The compiler must know the operative status and exceptions of a Traffic Regulation Order (TRO), Experimental Traffic Regulation Order (ETRO), Temporary TRO, and relevant redetermination order (RSO), as well as potentially applicable development-plan policies and approvals. Otherwise it may produce an illegal baseline or incorrectly discard viable proposals.

The compiler has two outputs. BASELINE RULES describe movements, times, vehicle classes and exemptions currently permitted on each graph edge or turn. PROPOSAL FLAGS identify new orders, redetermination, consents, heritage or equality review and potential policy tension. Only logically necessary physical, safety and explicitly approved scenario constraints narrow the candidate set. The agent may suggest interventions but cannot claim they are already lawful. Every output retains an identifiable, dated source and requires human confirmation.

Status tags refer to a 24-hour HackTheBurgh XIII prototype: CORE is the minimum demonstration; PRE-DEMO denotes offline preparation needed before a stable demonstration; STRETCH is optional. Because the event runs 7–8 November 2026, any uncompleted preparation is treated as a scope risk, not presumed complete. Cache inputs locally; do not make a successful live demo depend on council websites being available.

5.5.1 An Edinburgh order, end to end

Worked example — the City of Edinburgh Council's Lawnmarket Area Experimental Traffic Regulation Order 2026 (ETRO/25/23), advertised in February 2026, includes the following plain-English measure. This is an official advertised summary, NOT a verbatim operative legal clause; the signed order and schedules must be checked before it is enforced in a model:

> "Introduce an at any time prohibition of motor vehicles (with exemptions) on Johnston Ter, Lawnmarket, Castlehill and Upper Bow."
> — City of Edinburgh Council, ETRO/25/23 advertisement; measure (31) [official notice, 2026].

Read. The model identifies an advertised proposed/experimental movement rule, order reference ETRO/25/23, roads, vehicle class, "at any time" condition and unspecified exemptions. It does not infer the full exemptions, exact geographic schedule or operative amendments from the summary. These remain unresolved until the underlying order and schedules are inspected and approved by a human.

Bind. "Lawnmarket", "Castlehill", "Upper Bow" and "Johnston Terrace" are resolved against the chosen OpenStreetMap-derived graph, road-direction geometry and any signed order plans. The resolver emits candidate edge/turn IDs and a manual override queue. The language model selects among the offered IDs or abstains; it must not manufacture IDs or infer exemption lists.

```
RoadRule(
  rule_type  = "ACCESS_PROHIBITION",
  roads      = ["Lawnmarket", "Castlehill", "Upper Bow", "Johnston Terrace"],
  mode       = "motor_vehicle",
  schedule   = "all_day",
  exemptions = "VERIFY_SIGNED_ORDER",
  status     = "VERIFY_OPERATIVE_ORDER",
  source     = "Edinburgh ETRO/25/23, advertised measure 31",
  graph_ids  = "HUMAN_VALIDATED_LOOKUP"
)
```

Apply. Once the signed ETRO, schedules, exemptions and current status are verified, the baseline route graph can restrict applicable motor-vehicle movements while preserving exempt movements. Pedestrian and cycle movements require separate mode checks. A proposal to change that restriction remains a candidate but carries an APPROVAL_REQUIRED flag for a new or varied order. Scenario rules and road legality must not be confused with the binary physical edge-removal variable z.

COMPILED FIXINGS	a(e,m,t,s)=0 if a verified in-force order prohibits movement of mode m at time t in scenario s, subject to stated exemptions; proposal p -> {baseline_compatible, order_change_required, policy_tensions, unresolved}. Baseline legal movements and hypothetical physical changes are separate graph layers.

The defensible role of the language model is clause-level reading, source classification and location lookup—not deciding legal validity. Edinburgh's orders may amend, revoke or temporarily supersede earlier orders, so an unverified summary must be visibly marked unresolved. The output is a traceable planner aid rather than a substitute for checking consolidated instruments and obtaining the appropriate orders or consents.

5.5.2 Method and status

Stage	What happens	Status
Acquisition	Snapshot signed orders and amendments, advertised summaries, City Plan 2030/NPF4, council mobility material and published Old Town consultation evidence. Verify what is actually accessible.	PRE-EVENT
Street-to-edge resolver	Deterministic string and geometric matching from street names, directions and junction descriptions to edge and node identifiers on the OpenStreetMap-derived graph, with a manual override table.	PRE-EVENT

Stage	What happens	Status
Constraint schema	Types include baseline access, order status, exemption, approval trigger, policy assessment, designation and abstention; keep graph operations separate from legal movements.	PRE-EVENT
Triage	Per-section classification: compilable restriction, or not, or abstain.	CORE
Extraction	Schema-constrained decoding so malformed output cannot be produced; verbatim span first, structured fields second; literal-substring check on the span.	CORE
Verification	Quote supported? Source official? Signed order or summary? Operative on scenario date? Amendments checked? Unclear cases abstain.	CORE
Grounding	Model selects from the resolver's enumerated candidates or abstains.	CORE
Integration	Build verified scenario-specific legal baseline A_current and attach approval/policy flags to proposed changes; do not equate one-way restrictions with forbidden REMOVE_EDGE operations.	CORE
Abstention queue	Unresolved exemptions, road extents, stale/amended orders, ambiguous scope and planning judgement are surfaced for review, never asserted as legal facts.	CORE
Relaxation analysis	Relax compiled constraints one at a time and measure the change in the non-dominated set, identifying which provision costs the most benefit and in what terms.	OPTIONAL (if the loop is stable)

Grounding carries the greatest risk. An identifier that is generated rather than selected may be well-formed and wrong, binding a valid restriction to the wrong part of the network. That error is invisible in the output and propagates through every later evaluation, which is why the model is never allowed to emit an identifier directly.

5.5.3 Measuring extraction before we depend on it

Extraction must be tested rather than assumed accurate. Hand-check a small stratified sample against the official order and its schedules, with separate measures of clause extraction, street binding, time and class conditions, exemptions, status resolution and abstentions. OpenStreetMap restriction tags provide a secondary agreement check only; OSM can be stale or incomplete and is not the legal ground truth. Report denominators and limitations rather than presenting agreement as legal accuracy.

5.5.4 Points to settle before the build

Node-level operations. The search vector is z over edges plus continuous y. ADD_CROSSING and JUNCTION_REDESIGN act on nodes, and junction configuration is categorical, not binary, so constraints over

those two operations currently have nothing to bind to. Section 7 now adds node variables for both; if that proves too much for the day, the compiler is scoped to the edge operations and the limitation is stated explicitly. Either is defensible; the choice should be made deliberately, not met mid-build.

Junction configurations. "Split, merge or reconfigure junction behaviour" does not define admissible values. Section 7 now enumerates them (signalised, priority, roundabout, and banned-turn variants), which helps both the compiler and the junction model. PRE-EVENT.

Expected yield from the statutory development plan. City Plan 2030 and NPF4 provide policy tests, spatial designations and infrastructure priorities, but little text translates directly into literal edge-level prohibitions. NPF4 Policy 13 (sustainable transport), Policy 15 (local living), Policy 20 (blue and green infrastructure) and Policy 7 (historic assets and places) should primarily drive assessment criteria and safeguards. A low rate of automatic "hard constraint" compilation is the correct result when policy requires contextual judgement.

5.5.5 Strengthening the compiler

The method in 5.5.2 is the minimum that works. The following would make it materially better. Only the abstention queue (already in the table) is recommended for the event itself; the rest are tagged so that nobody attempts them on the day unless the main loop is finished.

Direction	What it adds	Status
Compiler inside the search loop	As described, compilation runs once. A retrieval formulation inverts this: for each candidate modification, retrieve the provisions bearing on the elements it touches and adjudicate that candidate. Inference then scales with the search rather than the corpus, handles rare provisions on demand, and keeps the local model busy throughout the optimisation rather than idle after a few minutes of batch reading.	OPTIONAL: only if the batch compiler and the search loop are both stable; the most consequential upgrade
Resolve the in-force position	TRO/ETRO/temporary and redetermination orders can supersede or vary earlier instruments. Inspect relevant signed order chains and effective dates; block legal-compliance claims when unresolved.	OPTIONAL, and if attempted then PRE-EVENT; not a hackathon-day task
Conditional and temporal restrictions	Examples: timed loading windows or access classes. Evaluate per mode, time and exception so mixed-use streets are not represented by one global binary value.	OPTIONAL
Permissions as well as prohibitions	A City Mobility Plan action or proposed street scheme creates a candidate-intervention prior, not permission to implement. Output a route to relevant legal review.	OPTIONAL
Measure against the manual alternative	Time-to-position comparison described in 5.5.3.	PRE-EVENT (estimate)

5.5.6 Further sources through the same pipeline

Acquisition, extraction, verification and grounding are source-agnostic. Once they exist, pointing them at more material is a matter of new schema entries, not new machinery. None of the following is proposed for the event; they are recorded so the first design does not foreclose them.

Source	What it yields	Status
City Plan 2030 land allocations and Delivery Programme	Future demand from adopted allocations, with sensitivity cases instead of assuming every approved site is already occupied.	OPTIONAL; PRE-EVENT if wanted for the demo
Edinburgh roadworks and temporary traffic-order information	Dated incidents and traffic-management changes when verifiable; model unknown periods with explicitly synthetic closures.	OPTIONAL
Cumulative development and tourism/event scenarios	Joint demand cases for allocations and Old Town pedestrian surges, with uncertainty ranges and no claim of calibrated forecasts.	OPTIONAL

Resident feedback is deliberately not a source for the compiler. A report of a dangerous crossing forbids nothing; it expresses a priority, and priorities enter through the objective (5.3), which the compiler does not touch. The two strands share acquisition tooling but differ in the work: consultation entries are short, unstructured and usually already carry coordinates, so the difficulty is deduplication and aggregation rather than reading legal prose. A fuller version that compiles resident and policy material into the objective itself, with a representation of admissible weightings and a position on how policy and resident priorities are reconciled, changes the formulation and should not be attempted alongside a first build (OPTIONAL, post-event).

6. Mathematical Core

The mathematical components are deliberately modular for a HackTheBurgh XIII prototype. A workable vertical slice uses a directed multi-modal graph, congestion-aware routes, a simple pedestrian-delay model, green-connectivity checks, a stochastic scenario generator, cited policy extraction and a small Pareto search. Legal status is a separate input to scenario feasibility, and the simulation is a relative comparison rather than a calibrated Edinburgh traffic forecast.

6.1 Spatial road graph

Represent the road system as a directed graph. Each junction, crossing or modal interface is a node. A road segment is a directed edge with length, capacity, free-flow travel time, lane count and operating characteristics.

GRAPH	$G=(V,E),\quad e=(u,v)\in E,\quad x_e=(\ell_e,c_e,t_e^0,n_e,s_e,\ldots)$ *Changing the city means changing graph topology G and/or the edge parameter vector x_e.*

6.2 Vehicle network flow

For an origin-destination demand matrix q_od, assign vehicle flow to the network. A standard equilibrium condition is Wardrop's principle: used routes between an origin and destination have equal minimum cost, and no unused route is cheaper.

EQUILIBRIUM	$f_r > 0 \Rightarrow T_r(f) = \min_{r' \in R_{od}} T_{r'}(f)$ *The route choice is endogenous: changing one road changes the flow pattern across the network. Under this equilibrium, adding a link can raise everyone's travel time (Braess's paradox), which is why HackTheBruh is allowed to remove roads as well as add them.*

Edge travel time can be made nonlinear to capture congestion. One practical approximation is a capacity-dependent function where travel time rises sharply as volume approaches capacity.

CONGESTION	$t_e(f_e) = t_e^0[1+\alpha(f_e/c_e)^\beta]$ *α and β control how aggressively congestion grows as utilisation approaches and exceeds practical capacity.*

6.3 Junction and signal modelling

A junction can be represented as a conflict graph. Turning movements become flows through conflict regions, and signal phases determine which movements can coexist. This supports optimisation of phase timing, pedestrian crossing windows and protected turning movements.

CAPACITY PROXY	$c_{m} = s_m \cdot g_m / C$ for movement m *s_m is saturation flow, g_m is effective green time and C is cycle length. The prototype can extend this with turn-specific conflicts and lost time.*

6.4 Pedestrian flow field

Rather than simulating every person individually from the start, treat pedestrians as a continuous density field. The model tracks how people accumulate, disperse and move through pavements, crossings and public-space corridors.

CONSERVATION LAW	$\partial\rho/\partial t + \nabla \cdot (\rho v) = 0$ *ρ(ξ,t) is pedestrian density and v(ξ,t) the local walking velocity at position ξ.*

Velocity can decrease with density, producing bottlenecks without having to hand-code every possible crowd interaction. For the prototype the equation is solved in one dimension along each pavement and crossing edge, so pedestrians live on the same graph as vehicles; a full two-dimensional field is only needed for open spaces such as squares.

CROWDING	$v(\rho) = v_0 \cdot \varphi(\rho), \quad \varphi'(\rho) < 0$ *As density increases, walking speed decreases. A simple monotone function is enough for the competition prototype; more advanced pedestrian models can be added later.*

6.5 Car-pedestrian interaction

The important planning insight is that road capacity and pedestrian capacity compete for space and time. A crossing can therefore be treated as a shared resource with conflicting flows and explicit safety or delay constraints.

SHARED RESOURCE	$g_{car} + g_{ped} \le C, \quad d_{ped} = h(\rho, g_{ped}, q_{car})$ *The signal cycle allocates time to competing movements; pedestrian delay depends on arrival rate, crowding and crossing opportunity.*

6.6 Green infrastructure as a graph

Green space is represented as a second spatial network. Patches are nodes and ecological or pedestrian connections are edges. This lets HackTheBruh measure whether green infrastructure is fragmented or well connected.

SPECTRAL CONNECTIVITY	$L = Deg - A, \quad \lambda_2(L) \uparrow \Rightarrow \text{connectivity} \uparrow$ *Deg is the degree matrix and A the weighted adjacency matrix of the green graph. The second-smallest Laplacian eigenvalue λ_2 measures how hard the graph is to cut apart: higher means a better connected network.*

One caveat matters in practice: λ_2 is exactly zero for every disconnected graph, so on its own it cannot tell a network in two pieces from one in ten. Real green networks usually start fragmented, so HackTheBruh pairs λ_2 with the number of connected components (the multiplicity of the zero eigenvalue) and the share of green area in the largest component. Those two give the optimiser a usable signal while the network is still fragmented; λ_2 takes over once it is joined up. Other metrics can be added, including total green area, average distance to green space, corridor continuity and a penalty for cutting existing green links.

6.7 Emissions and cost proxies

A full emissions model is unnecessary for a prototype. A useful first model is an edge-level proxy that increases with time spent, stop-start behaviour and flow concentration. The important requirement is consistency rather than false precision.

EMISSION PROXY	$E = \Sigma_e [a \cdot f_e \cdot t_e + b \cdot idle_e + c \cdot stop_e]$ *A calibrated version could later use vehicle classes and engine-specific emission factors. For a hackathon prototype, this is an interpretable comparative score.*

6.8 Multi-objective optimisation

Urban design has no single objective. Instead, HackTheBruh optimises a vector of outcomes and keeps the non-dominated designs. This is the Pareto frontier.

PARETO PROBLEM	min_x F(x)=[J_car,J_ped,J_green,J_emit,J_voice,J_cost,J_risk] *Every entry is written as a cost to be minimised, so J_green is the negative of green connectivity. A candidate is Pareto-dominated if another candidate is no worse on every objective and strictly better on at least one.*

The interface can let the user change weights or select hard constraints. The system then shows which candidate regions become feasible rather than silently collapsing all policy choices into one score.

6.9 Robust optimisation under uncertainty

Demand is not fixed. The system can sample normal and abnormal conditions: commuting peaks, school arrivals, stadium events, incidents, weather-related shifts and temporary closures. Rather than optimising only the mean result, HackTheBruh explicitly penalises the tail of the loss distribution.

RISK-AWARE OBJECTIVE	J_robust(x)=E[J(x,D)] + λ·CVaR_α(J(x,D)) *CVaR_α is the average loss over the worst (1−α) fraction of scenarios, for example the worst 5% when α = 0.95. This makes the optimiser prefer designs that are less fragile to disruptions.*

7. City Topology Optimisation

The distinctive feature of HackTheBruh is that it is allowed to change the graph itself. The search space can include adding or removing road links, splitting junctions, moving crossings, changing lane allocation, creating pedestrian-only corridors, or inserting green connectors.

Operation	Meaning
ADD_EDGE	Create a new road or pedestrian connector where geometry permits.
REMOVE_EDGE	Remove or convert a road segment when it causes disproportionate congestion, danger or fragmentation.
REPARTITION	Change lane allocations between through traffic, turning traffic, buses, bikes or pedestrians.
ADD_CROSSING	Introduce or relocate a pedestrian crossing and alter signal timing.
GREEN_CORRIDOR	Convert a sequence of parcels or edges into a connected green corridor.
JUNCTION_REDESIGN	Split, merge or reconfigure junction behaviour.
HYBRID SEARCH	x=(z,y), z∈{0,1}^|E∪E_cand|, y∈R^k *Binary variables z switch existing edges E and candidate new links E_cand on or off; node variables (below) cover crossings and junction type; continuous variables y can represent timings, widths or signal splits. This creates a mixed discrete-continuous optimisation problem.*

Exact global optimisation of arbitrary graph topology is computationally expensive. The practical approach is a hierarchical search: use cheap heuristics to generate candidate modifications, use GPU simulation to evaluate them, and apply local improvement or evolutionary selection to the strongest candidates.

Two of the six operations act on nodes rather than edges, and the encoding above has no variable for them. HackTheBruh therefore adds a binary crossing indicator per candidate node and a categorical junction-type variable, so that compiled constraints over ADD_CROSSING and JUNCTION_REDESIGN have something to bind to. Enumerating the admissible junction types is PRE-EVENT work; if node variables prove too much for the day, the compiler is scoped to edge operations and says so.

NODE VARIABLES	$x = (z, y, w, k)$, $w_v \in \{0,1\}$ (crossing at node v), $k_v \in K = \{$signalised, priority, roundabout, signalised-with-banned-turns, ...$\}$ *w fixes whether a crossing exists at a candidate node; k picks a junction configuration from an enumerated set K. Both can be fixed by the constraint compiler in the same way as z.*

8. From Planning to Real-Time Urban Operations

The mathematical core can also explore operational changes after a scheme is implemented: illustrative signal timing, pedestrian priority, route management and incident responses. HackTheBruh makes recommendations only. Real-life intervention requires authority, engineering validation, safety assurance, and where applicable statutory orders and operational integration.

RECEDING HORIZON	$u_t^* = \text{argmin}_u \ \Sigma_{\tau=t}^{t+H} J(x_\tau,u_\tau)$ subject to dynamics *At each step, optimise the next H time units, apply the first control action, observe the new state and repeat.*

This is a model-predictive-control interpretation of the urban agent. The agent can decide when to re-optimise, which scenario family to test, and whether a detected change is large enough to justify intervention.

9. Data and City Representation

The HackTheBurgh XIII prototype should avoid restricted feeds and proprietary datasets. OpenStreetMap geometry, licensed Edinburgh Council spatial layers, published council documents and explicitly synthetic origin-destination demand are enough to demonstrate the workflow. Any operational conditions not verified from source documents should be labelled assumed.

Static geometry: Road centre lines, junctions, crossings, paths, parks, green corridors and land-use proxies.

Demand: OD matrices generated from population/attractor proxies or imported from public datasets when available.

Operational parameters: Speed limits, lane counts, capacity estimates, signal timings or synthetic defaults.

Scenario generator: Rules for peaks, closures, demand multipliers, event surges and pedestrian concentration.

Policy corpus: Edinburgh TROs/ETROs/temporary orders and schedules; relevant redetermination orders; NPF4; adopted City Plan 2030 written statement and interactive proposals map; City Mobility Plan, Implementation Plan and Our Future Streets. Store source URL, effective date, scope, version and a reference to the actual operative document.

Public feedback: Published City of Edinburgh Council Consultation Hub responses and Old Town Streets material, plus attributable published submissions by local accessibility, walking, cycling, heritage and business groups; record aggregate themes and usable locations without personal data.

10. Hackathon Demonstration

The HackTheBurgh XIII demonstration uses a compact, map-bounded Old Town study area centred on Lawnmarket and Castlehill, with nearby approaches such as Johnston Terrace and Upper Bow. The council's Lawnmarket Area experimental traffic order ETRO/25/23 supplies a real example of mode, timing and exemption handling. The demo does not claim to reconstruct every city-centre order or predict measured traffic outcomes; synthetic demand and assumptions are labelled.

Demo flow

1. **Load the Edinburgh Old Town road and footway graph, show a clearly marked synthetic-demand baseline, and highlight the pedestrian-priority area.**
2. **Load cached Edinburgh policy and road-order records. Display a verified excerpt from ETRO/25/23, distinguish advertised summaries from signed operative provisions, and show unknown exemptions or order status as a visible abstention/verification task.**
3. **Ask the agent: "Test ways to reduce walking delay on Lawnmarket without losing the essential permitted servicing window, and show network, heritage, accessibility and legal-order trade-offs."**
4. **Let a human approve the scenario assumptions and record classification; generate a small set of candidate crossings, timing, lane/space allocations or route changes.**
5. **Evaluate batched candidates under peak visitor flows, delivery/servicing windows, local incidents and uncertain synthetic demand; accelerate with a GPU only if available. City Plan 2030 allocations are an optional future-demand sensitivity case.**
6. **Show a Pareto comparison across pedestrian delay, vehicle diversion, servicing-access preservation, green/heritage safeguards, public concerns, cost and robustness.**
7. **Open an option to inspect the precise graph and signal changes, baseline restrictions, assumptions, and the roads or turns affected.**
8. **Open a separate legal and policy panel: current order compatibility, any new TRO/RSO or other approvals to investigate, NPF4/City Plan 2030 considerations and published consultation themes.**
9. **Test a Johnston Terrace closure or an Old Town visitor-demand surge as an explicitly synthetic disruption, comparing it with the baseline.**
10. **Change the weight on pedestrian accessibility versus vehicle diversion, and observe how the preferred non-dominated options change without treating strategy as law.**

Steps 8 and 10, and the future-demand scenario in step 5, rely on stretch items in Section 12. The demo is designed so that it still holds together if any of them is cut.

11. Evaluation Framework

HackTheBruh should be evaluated quantitatively and with explicit epistemic limits. Compare options on a common synthetic demand scenario, measure quote and mapping accuracy on hand-checked samples, show unresolved legal status rather than guessing, and distinguish mathematical performance from statutory approval.

Metric	Definition	Interpretation
Total vehicle travel time	Σ_e f_e·t_e	Lower is better, subject to feasibility.
95th percentile delay	quantile_{0.95}(delay)	Captures severe user-facing congestion.
Pedestrian accessibility	mean or percentile path time	Measures walking performance across the district.
Green connectivity	λ_2(L_green) or corridor continuity	Measures structural connectivity rather than area alone.
Robustness	CVaRα(J)	Measures performance in adverse scenarios.
Baseline legality and approval flags	verified operative conditions satisfied; proposals needing review	Report verified legal-baseline restrictions separately from planning-policy tensions and orders/consents required for hypothetical designs.
Public-voice improvement	1 − J_voice(x)/J_voice(baseline)	How much a design improves conditions where concerns were reported.
Extraction accuracy	precision on a hand-checked sample	Share of extracted clauses and concerns a human agrees with; quote verification must be 100%.
Compiler agreement	match rate vs held-out OSM restriction tags	Recompiled restrictions compared with the map attribution they were hidden from; reported with the number of abstentions.
Computational throughput	candidate evaluations / second	Reports candidate/scenario throughput on the actual available CPU or GPU, without a sponsor-hardware claim.

12. One-Day Competition Scope

The full architecture exceeds the HackTheBurgh XIII build window (7–8 November 2026). The right strategy is a reproducible, end-to-end Edinburgh Old Town vertical slice with cached legal evidence and scenario simulations; defer comprehensive legal-status reconciliation and calibrated traffic forecasting.

Scope	Recommended content
Must have	Graph representation; synthetic or imported OD demand; congestion-aware vehicle routing; one edge-based pedestrian model; green-connectivity metric; stochastic scenario generator with batched evaluation on the GPU; candidate search over a small set of operations; local open model with tool calling; policy and consultation ingestion with quoted, cited extraction; constraint compiler over the district's TROs feeding fixings into the search, with an abstention queue; public-concern heatmap; map view with a simple Pareto scatter plot.
Strong stretch	Lane allocation decisions; signal optimisation; interactive Pareto UI with live re-weighting; CVaR-based robustness; automatic agent critique loop; clause-by-clause policy check; node variables for crossings and junction type; constraint relaxation analysis; future-demand scenarios from emerging-plan site allocations.
Extreme stretch	Topology-aware mixed-integer optimisation, differentiable simulation, live sensor ingestion, learned surrogate models, an RL controller for short-horizon signal control, the compiler inside the search loop, or in-force resolution of amended orders.

Build plan for the policy and public-voice layer

The policy/road-order subsystem can be built in parallel with simulation and connected through a fixed JSON interface: baseline access rules, approval flags, policy evidence, scenario priors and located concerns. No corpus claim should exceed the documents actually verified.

Step	What we build	Fallback if time runs short
1. Corpus (before the event)	Collect the signed or available ETRO/25/23 documents and schedules, relevant amended TROs/RSOs, NPF4 and City Plan 2030 extracts, mobility plan and public Old Town consultation summaries; preserve reference and version.	Use one verified order extract, NPF4 Policies 7 and 13, relevant City Plan 2030 provisions and one published consultation summary.
2. Schema and prompts	Agree one schema for source type, legal status, effective date, street, mode, exemptions, quote, graph binding, approval flags and abstention; create a small gold sample.	Drop optional fields; keep source, quote, type, location.
3. Retrieval	Embed sections locally; filter by district street names and transport keywords before extraction so the model only reads relevant passages.	Keyword filtering only.
4. Grounding	Match extracted street and junction names to OpenStreetMap names on the graph; fuzzy match with a manual override table.	Hand-map the dozen locations that matter for the demo district.

Step	What we build	Fallback if time runs short
5. Compile	Build three distinct outputs: a scenario access mask from checked order restrictions; “approval required” flags for proposals; and non-binding policy/strategy priors.	Show one verified baseline access rule and a simulated change explicitly labelled “would require legal review”.
6. Concern field	Compute s_e^j, add J_voice to the objective vector, render the heatmap layer.	Heatmap only, with J_voice reported but not optimised.
7. Verify and explain	Exact quoted-span check; status and order-review checklist; human confirmation; four-part design panel (baseline restrictions, proposed approvals, policy assessment, resident concerns).	Quote check plus a static approval list.

A stratified sample of around thirty records, if sufficient material is available, gives useful preliminary extraction agreement with human review; record the source type, sample size and abstentions. Cache the extracted output for reliable judging. Do not call a small demo sample legal-compliance validation.

Pre-event work (team repository)

These are pre-demo readiness tasks. If one is incomplete, narrow the source and graph extent and disclose the limitation rather than presenting the information as fully reconciled.

Item	Definition of done	Owner
Document snapshots	Snapshot official Edinburgh order, schedules/advert, NPF4, City Plan 2030, mobility policy and one consultation source relevant to the Old Town, with citations and effective dates.	
Street-to-edge resolver	Function from (street name, direction, optional junction) to candidate edge/node ids on the OSM graph, with fuzzy matching and a manual override table; tested on the demo district.	
Constraint schema	Typed baseline restriction and proposal approval schema; separate law/policy/strategy/public-feedback classes, abstention and exemptions.	
Junction type enumeration	The set K of admissible junction configurations, agreed with whoever builds the junction model.	
Extraction agreement test	Evaluate against a small hand-labelled set of signed-order clauses (where accessible) and OSM tags as secondary evidence; report abstentions honestly.	

Item	Definition of done	Owner
Manual-effort estimate	An honest estimate of how long assembling the district's restrictions by hand would take a planner, for the comparison on the day.	
Cached extraction output	Cache validated Old Town records and synthetic demand; no live dependency during judging.	

13. Suggested Technical Stack

Python: Experiment orchestration, data processing and agent tool APIs.

C++ / CUDA: Only if needed. The first choice is to batch scenarios as tensor operations in PyTorch or CuPy, which is achievable in a day; custom kernels are a later optimisation.

PyTorch: Open-model inference, learned surrogates and future differentiable components.

Local or licensed LLM with structured output: quotation-backed policy extraction, agent tool calling and explanations. Run on a laptop or event infrastructure; cache results, and use cloud inference only within document/data permissions and competition rules.

PyMuPDF / pdfplumber, local embedding model, FAISS: Document parsing and retrieval over the policy and feedback corpus.

OSMnx and fuzzy name matching: Grounding extracted street and junction names onto the city graph.

NetworkX / SciPy: Graph algorithms, sparse linear algebra and rapid prototyping.

FastAPI or local service: Stable tool interface between the agent and the mathematical engines.

MapLibre / Deck.gl or lightweight WebGL: Interactive urban map, candidate comparison and heatmaps.

14. Risks and Mitigations

Simulation realism: Use labelled synthetic OD demand, relative comparisons and sensitivity analysis, not precise Edinburgh travel-time forecasts without calibration.

Combinatorial explosion: Use hierarchical search, candidate pruning, stochastic sampling and surrogate models. Let the agent allocate compute to promising regions.

Agent hallucination: All numerical claims must originate from tool outputs. The agent may propose actions but may not invent simulation results.

Misread policy: NPF4/City Plan 2030 require contextual planning judgement; verify verbatim spans and keep them as policy checks unless a human identifies a justified binding scenario constraint. A legal order's effective dates and exceptions need separate verification.

Wrong binding: Bind restrictions only to resolver-enumerated roads/turns, with manual checks against order plans and an abstention queue. Do not infer geometry from a road name alone.

Stale orders: In-force status, amendment chains and experimental order variation may be uncertain. Mark every incomplete chain UNRESOLVED and never claim the resulting network is a verified legal representation.

Unrepresentative feedback: Consultation responses over-represent the motivated. Keep public voice as a separate, visible objective, show comment counts, and never present it as a survey of the whole city.

Terms of use and privacy: Respect source-specific licensing, remove personally identifying consultation data, and document whether inference is local or remote before uploading any material.

Compute and inference limits: The software must run on ordinary event hardware. Prefer cached document extraction, CPU-friendly flow algorithms and bounded candidate batches; use available GPU acceleration opportunistically rather than depending on any particular sponsor machine.

Over-optimisation: Use constraints and robustness penalties to stop the optimiser finding mathematically cheap but operationally absurd solutions.

Data gaps: Use synthetic demand and clearly labelled assumptions where public data is incomplete.

15. Research Extensions

Differentiable city simulation: Construct smooth approximations of routing and congestion so selected parameters can be optimised with gradient methods.

Graph neural networks: Learn fast surrogate models $\hat{J}(G,x,D)$ from previous simulations so the optimiser does not need to simulate every candidate at full fidelity.

Multi-agent planning: Use specialist agents for transport, pedestrians, environment and resilience, with a coordinator resolving conflicts.

Scheme screening: Run the policy check in reverse: take a proposed scheme or planning application and report how it performs against policy and reported concerns before it goes to consultation.

Closing the consultation loop: Re-ingest feedback after a scheme is built to test whether the concerns that motivated it have actually faded.

Online Bayesian optimisation: Choose the next simulation to maximise expected improvement or information gain rather than sampling uniformly.

Digital twin integration: Replace synthetic state with live traffic counts, signal state and sensor streams where available.

Microscopic validation: Run final candidates through a higher-fidelity simulator only after the coarse optimiser has narrowed the search space.

Appendix A — Mathematical Formulation

This appendix collects a compact mathematical specification that can be used as the technical backbone of the prototype. The exact functions can be simplified during implementation; the important point is that each component has a clear place in the model.

Component	Formulation	Meaning
Decision variables	G, x, u	G = selected graph topology; x = physical/design parameters; u = operational controls.
Demand uncertainty	$D \sim P(D \mid z)$	z can encode time-of-day, incidents, events and other latent scenario conditions.

Component	Formulation	Meaning
Vehicle flow conservation	$\Sigma_{e\in\delta^+(v)} f_e - \Sigma_{e\in\delta^-(v)} f_e = b_v$	For every node and each origin-destination commodity, net outflow equals that node's demand balance b_v.
Travel time	$t_e=t_e(f_e,c_e,t_e^0)$	Any monotone congestion function can be used; nonlinear growth is desirable.
Path cost	$T_r=\Sigma_{e\in r} t_e$	The cost of a route is the sum of edge travel times plus optional penalties.
Pedestrian PDE	$\partial\rho/\partial t + \nabla\cdot(\rho v)=q$	q is the source/sink term for people entering or leaving the modelled region.
Pedestrian speed	$v=v_0\,\varphi(\rho)$	φ decreases as density increases.
Green connectivity	$L=Deg-A$; $\kappa_{green}=\lambda_2(L)$	Higher λ_2 generally means a better connected green graph.
Robust objective	$J_{robust}=E_D[J] + \lambda\,CVaR_\alpha(J)$	Balances average performance against tail risk.
Multi-objective form	$F=[J_{car},J_{ped},J_{green},J_{emit},J_{voice},J_{cost},J_{risk}]$	The optimisation seeks non-dominated candidates.
Topology constraint	$G\in\mathscr{G}_{feasible}$	Physical and safety feasibility plus reviewer-approved constraints; legal baseline and proposal-order requirements are separate.
Budget constraint	$\Sigma_i\,cost_i\,z_i \le B$	Limits the amount of infrastructure change.
Receding-horizon control	$u_t^*=\operatorname{argmin}_u \Sigma_{\tau=t}^{t+H} J(x_\tau,u_\tau)$	Used for real-time signal and routing control.
Surrogate model	$\hat{J}=NN_\theta(G,x,D)$	A learned approximation can accelerate search after enough simulations have been collected.
Bayesian acquisition	$x_{next}=\operatorname{argmax}_x \alpha(x \mid data)$	Select the next expensive simulation using expected improvement or information gain.
Policy feasibility	$A_{current}(s)$ = {movements permitted by verified in-force orders in scenario s}; proposals p carry review_flags(p).	Do not equate plan-policy language with statutory road prohibitions or treat model feasibility as official permission.
Compiled fixings	access(e,m,t,s) in {permitted, prohibited, unresolved}; z_e controls hypothetical geometry separately.	Apply approved road restrictions to the scenario access layer; flag changes requiring TRO/RSO/other action.

Component	Formulation	Meaning
Node variables	w_v∈{0,1}, k_v∈K	Crossing indicator and categorical junction type, so node-level constraints have something to bind to.
Concern field	s_e^j=Σ_i sev_i·K(dist(e,loc_i))·1[theme_i=j]	Located public concerns spread over nearby edges by theme.
Public-voice objective	J_voice=Σ_j Σ_e s_e^j·m_e^j(x)	Rewards improving the matched metric where concerns were raised.

Appendix B — The Competition Narrative

HackTheBruh is intentionally positioned as more than a traffic simulator. The simulator provides the physics-like computational layer; the optimiser searches the design space; the language model reads the city's policies and its residents' feedback, turns them and the planner's own language into machine-executable constraints and objectives, and decides which experiments to perform; the interface makes the resulting trade-offs visible.

One-sentence pitch: "HackTheBruh is an Edinburgh-focused agentic urban-operations compiler that translates Scottish planning policy, road-order evidence and public feedback into source-traceable street-design experiments and robust, explainable trade-offs."

Core differentiator: The system does not ask an AI to guess how a city should work. It asks the AI to orchestrate a computational search over thousands of candidate cities and present the human planner with transparent, robust trade-offs.

Why it belongs on the urban track: The inputs are the ones a real council already has and struggles to use together: its own policy documents and what its residents have told it. HackTheBruh connects that evidence to a rigorous model of the network, so a planner can show not only that a design performs well, but that it respects the restrictions actually in force, with a citation for each, and that it answers what people asked for.

Appendix C — Edinburgh Source Register and Verification Notes

Evidence snapshot: 7 November 2026. These are retrieval starting points, not a consolidated legal dataset. Exact local provisions must be verified against original documents, maps, operative status and exemptions before use. This document is a technical concept, not legal advice.

01. Competition — HackTheBurgh XIII official website; 7–8 November 2026, The Nucleus, Edinburgh. https://hacktheburgh.com/

02. Statutory planning — City Plan 2030: adopted 7 November 2024; includes written statement, Appendix D and proposals maps. https://www.edinburgh.gov.uk/cityplan2030

03. National policy — Scottish Government: National Planning Framework 4, published 13 February 2023. https://www.gov.scot/publications/national-planning-framework-4/

04. Planning interpretation — Scottish Chief Planner: NPF4 transitional arrangements and planning judgement, February 2023. https://www.gov.scot/publications/chief-planner-letter-transitional-arrangements-for-national-planning-framework-4/

05. Mobility — City of Edinburgh Council: City Mobility Plan, approved 2021 and updated February 2024. https://www.edinburgh.gov.uk/cmp

06. Road orders — City of Edinburgh Council: View or comment on traffic orders; includes order types and portal links. https://www.edinburgh.gov.uk/trafficorders

07. Edinburgh worked example — Council: Lawnmarket ETRO/25/23 official published advertisement; its measures are summaries and require review of order schedules. https://www.edinburgh.gov.uk/downloads/file/38911/advert

08. Lawnmarket council project — Council: Lawnmarket project, pedestrian and cycle arrangements and experimental trial information. https://www.edinburgh.gov.uk/public-transport/lawnmarket-restricting-vehicle-access/2

09. Old Town project — Council: Improving Old Town Streets, scope and public consultation history. https://www.edinburgh.gov.uk/public-transport/improving-old-town-streets

10. Road redetermination — Roads (Scotland) Act 1984, section 152(2), redetermination of means of public passage. https://www.legislation.gov.uk/ukpga/1984/54/section/152

11. Spatial data — Edinburgh Council Open Spatial Data Portal, dataset-specific licence requirements. https://data.edinburghcouncilmaps.info/

12. Consultation — City of Edinburgh Council Citizen Space consultation hub and published outcomes. https://consultationhub.edinburgh.gov.uk/

Legal and implementation safeguards for the HackTheBurgh XIII demonstration

- The public notice for ETRO/25/23 is not the signed operative text. Confirm scope, order commencement, variations, exceptions, access windows and current status before applying any baseline rule.
- Maintain distinct record types for in-force road orders, proposed orders, redetermination orders, statutory development-plan policy, non-statutory mobility strategy and published public comments.
- For proposals altering use of a road or public passage, report “approval/order review required”; never state that an optimiser may lawfully implement the change.
- Treat NPF4 Policy 7, 13, 15 and 20 as policy evidence requiring judgement. City Plan 2030 mapping layers and historic designations require spatial cross-checking.
- Where a rule cannot be verified, abstain and provide a clearly labelled illustrative simulation. Use explicit data vintages and only publish measurable quantities actually produced by the model.